

Beyond the Keyboard

HCI – Human Computer Interaction

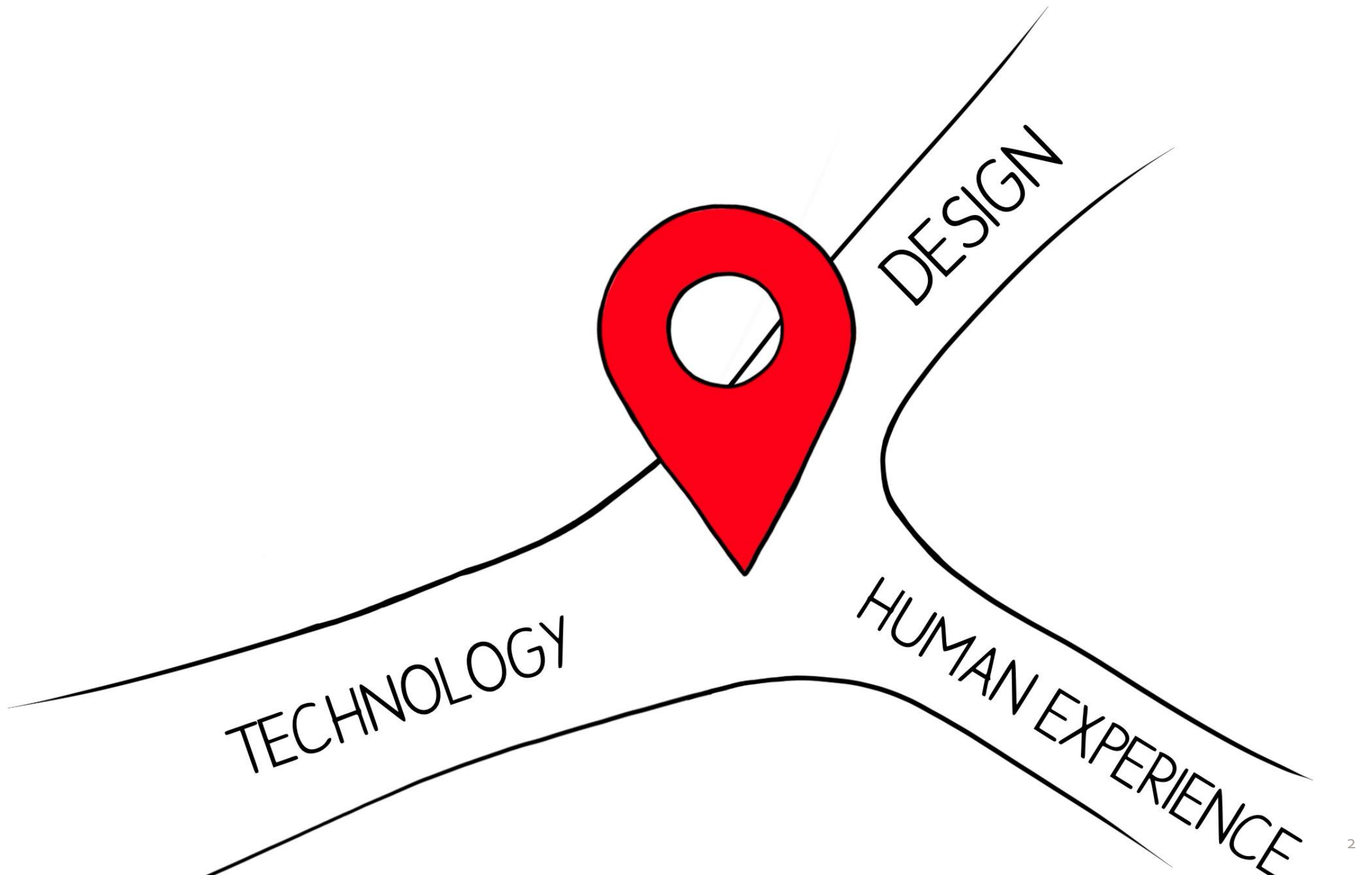
Modern Human Computer Interfaces



Prof. Dr.-Ing. Biying Fu
Explainable AI
RheinMain University of Applied Sciences

Author's Note: unless indicated otherwise, all sketched figures and icons are hand-drawn by myself and licensed as Creative Commons Attribution-ShareAlike 4.0 International License







Biying Fu

Since 06/2022 Professor for **Applied Informatics with Major in Explainable AI** in the department of Design, Computer Science, and Media at RheinMain University of Applied Science.

“

Human-Computer Interaction (HCI) is a discipline concerned with the design, evaluation and implementation of interactive computing systems for human use and with the study of major phenomena surrounding them.

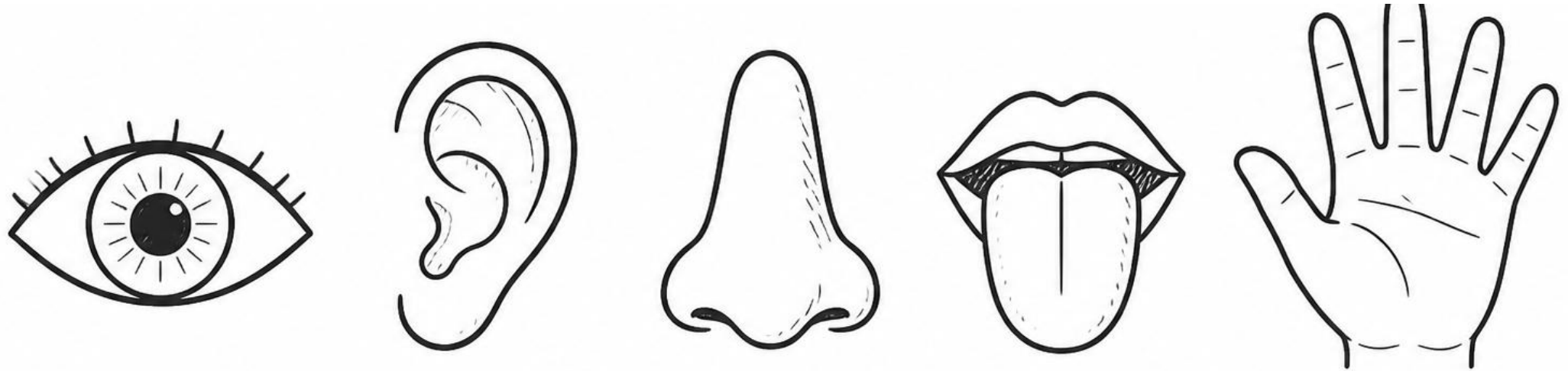
”

Alternative terms: Human-Machine Interaction (HMI), Human-Machine Communication, Man-Machine Interaction/Interface, or Ergonomics



Interaction Interfaces

- How to perceive the world? – sense

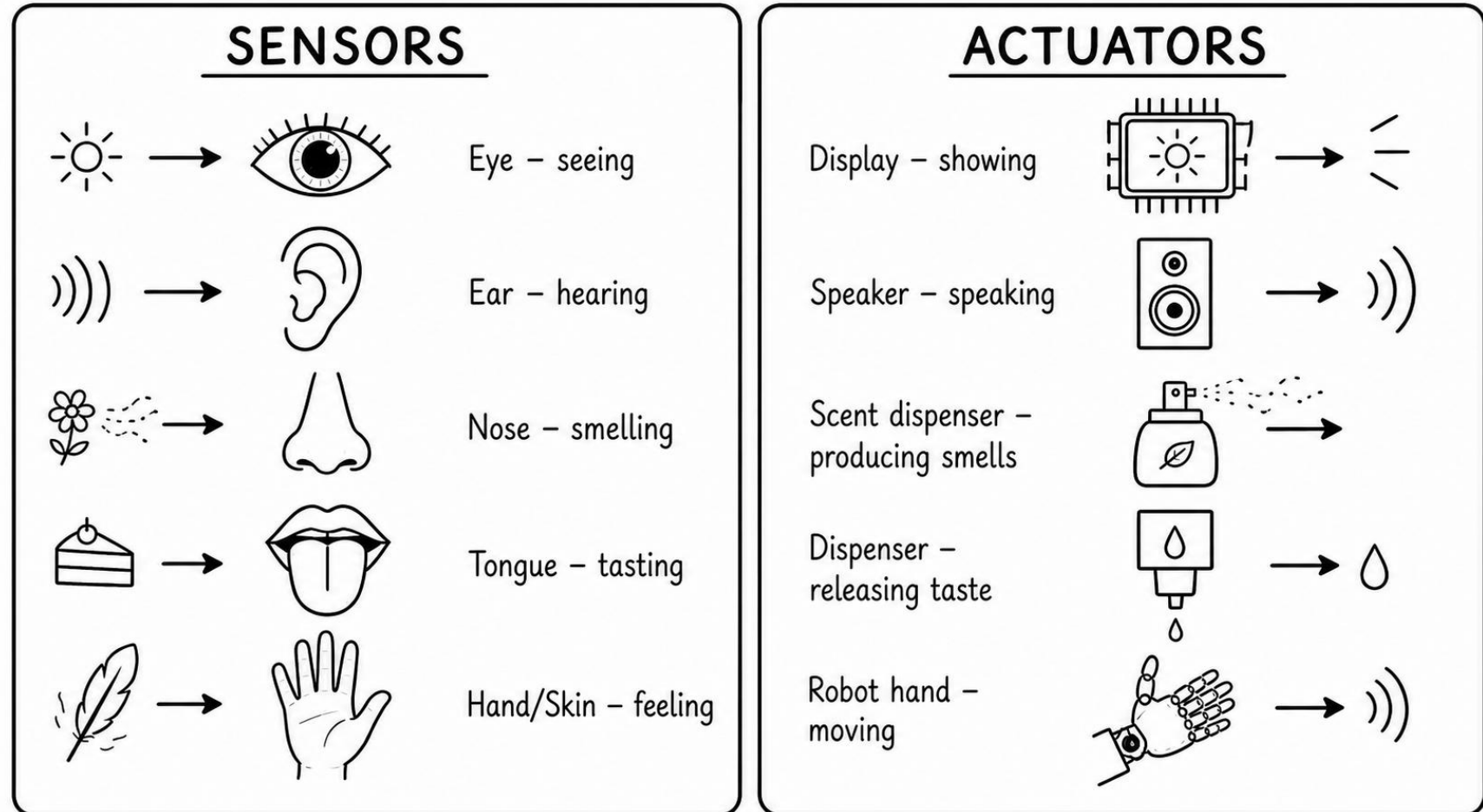


Images generated with GPT 5.6 Luna



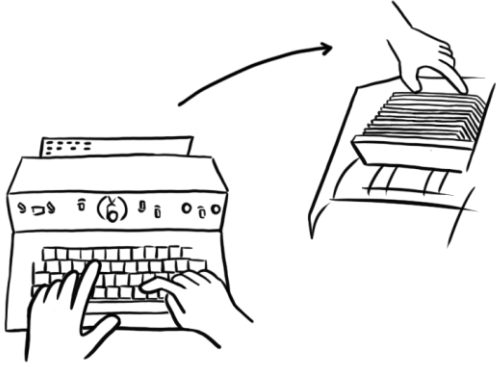
Interaction Interfaces

How to interact with the world? – act



Images generated with GPT 5.6 Luna

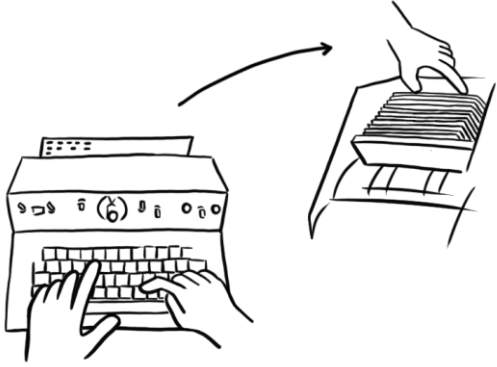
Interfaces are Evolving!



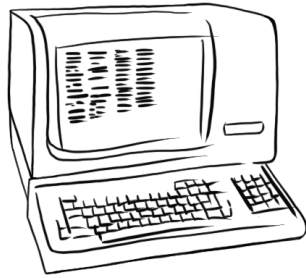
*No or delayed
interaction, e.g. with
punch cards*



Interfaces are Evolving!



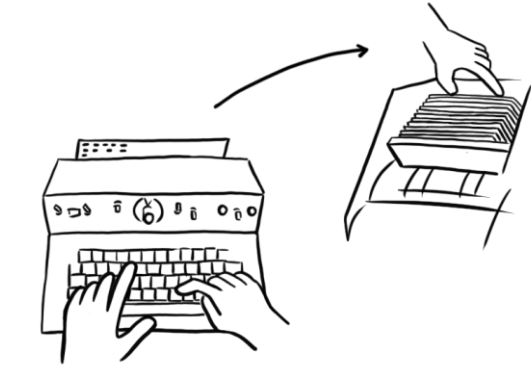
*No or delayed
interaction, e.g. with
punch cards*



*Command Line Interface
(CLI)*



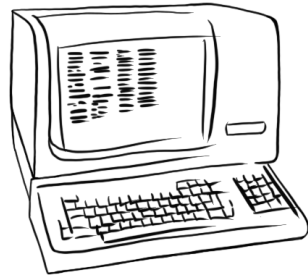
Interfaces are Evolving!



*No or delayed
interaction, e.g. with
punch cards*



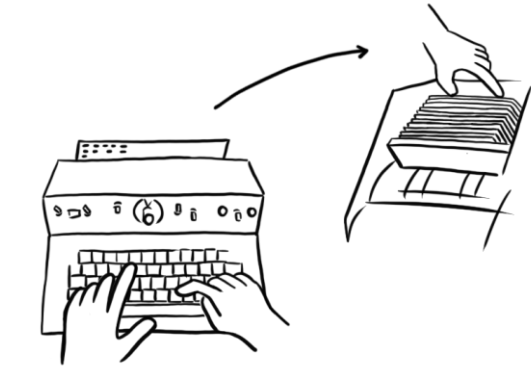
*Graphical User
Interfaces (GUI)*



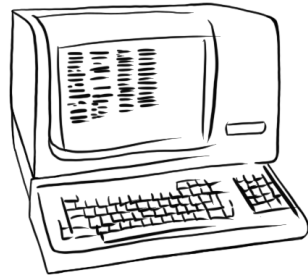
*Command Line Interface
(CLI)*



Interfaces are Evolving!



*No or delayed
interaction, e.g. with
punch cards*



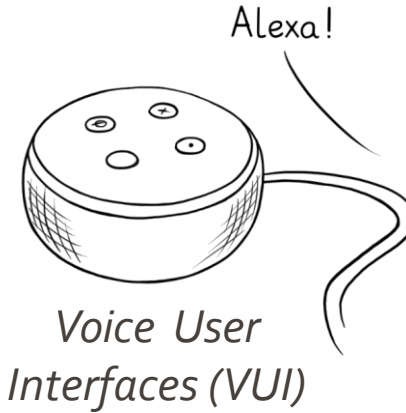
*Command Line Interface
(CLI)*



*Graphical User
Interfaces (GUI)*



Direct Touch

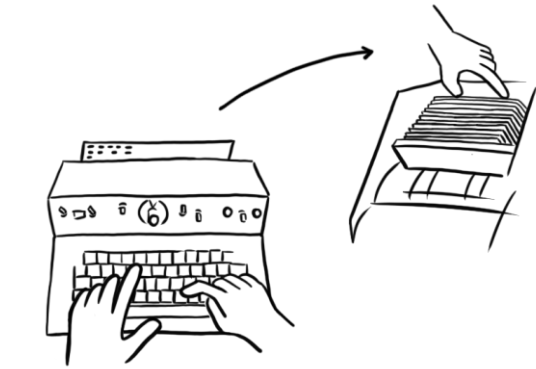


Alexa!

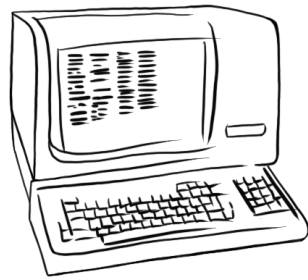
*Voice User
Interfaces (VUI)*



Interfaces are Evolving!



*No or delayed
interaction, e.g. with
punch cards*



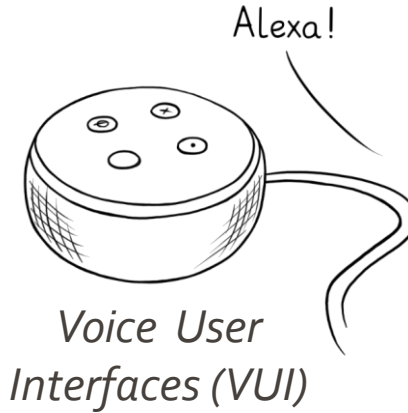
*Command Line Interface
(CLI)*



*Graphical User
Interfaces (GUI)*



Direct Touch



*Voice User
Interfaces (VUI)*

*Spatial Computing
AR/VR*

*Wearable and
Embodied Computing*

*Multimodal
Computing*

TODAY

TIME

What is a
computer?



IBM PC 5150

International Business Machines Corp., Armonk, 1981

Im August 1981 präsentierte IBM seinen ersten PC.
Der Rechner wurde unmittelbar ein Erfolg.

Dieser PC kam von der Computerfirma schlechthin.
Das schaffte auch bei Firmen Zuversicht, die den
bisher auf dem Markt erhältlichen Modellen nicht trauten.
Der Personal Computer eroberte endlich die Büros.

Der PC, der mit Microsofts Betriebssystem DOS lief,
setzte nun den lang fehlenden Industriestandard.
Zahlreiche «Klone», Kopien, die ebenfalls auf DOS
und dem Intel-Prozessor basierten, folgten dem IBM-PC.

Computer Museum Paderborn (2020)
Image: Marion Koelle



Apple Watch (2024)
Image: Shutterstock

Today, Computing is Ubiquitous.



Ubiquitous Computing: The most profound technologies are those that disappear. They weave themselves into the fabric of everyday life until they are indistinguishable from it.

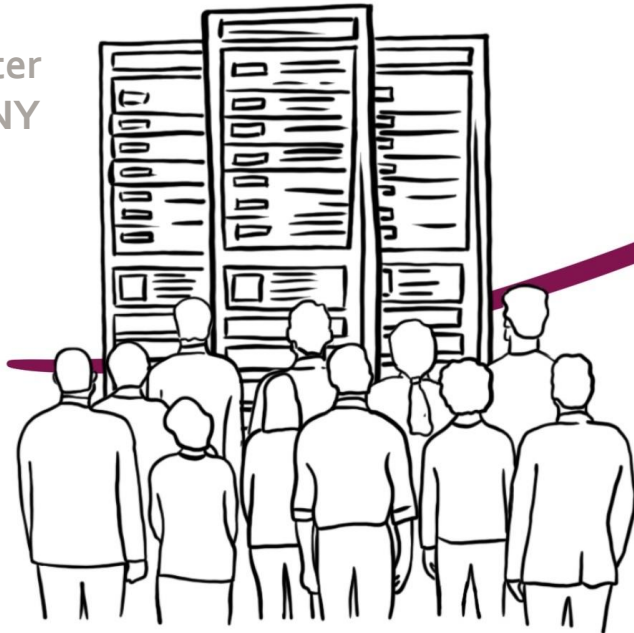
Mark Weiser (1998)



Today, Computing is Ubiquitous.

**NUMBER
of devices**

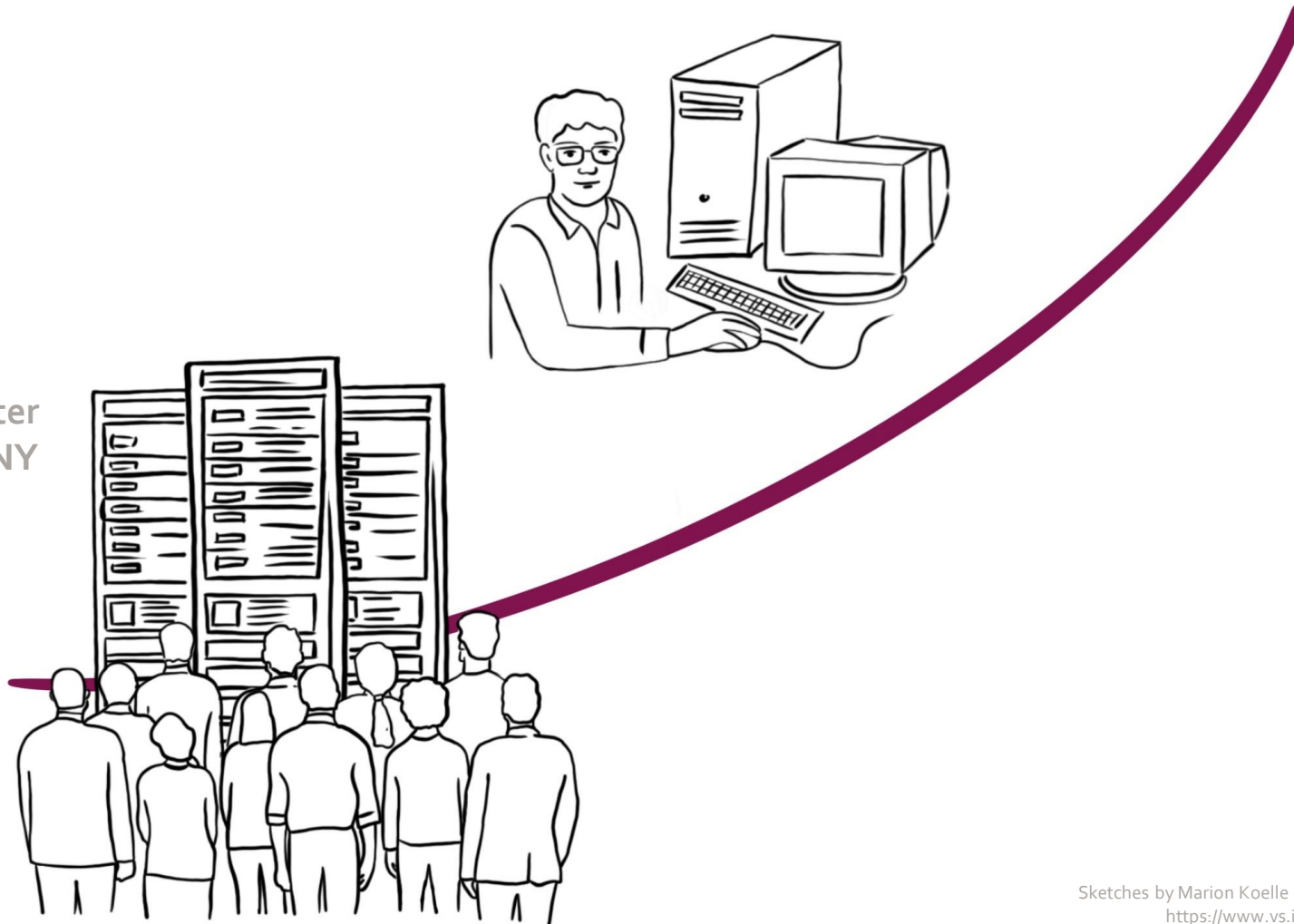
ONE computer
for MANY



Today, Computing is Ubiquitous.

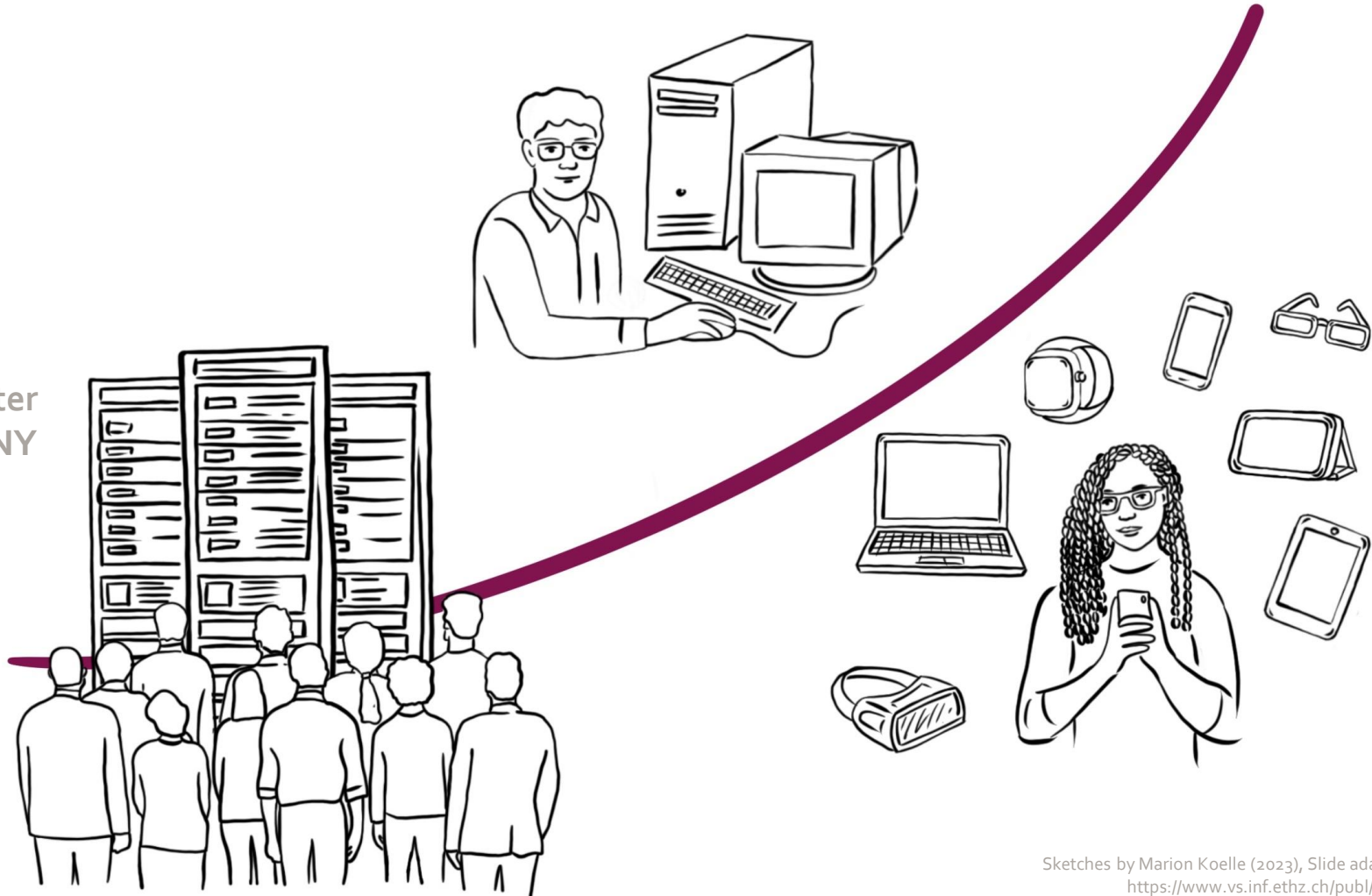
**NUMBER
of devices**

ONE computer
for MANY



Today, Computing is Ubiquitous.

ONE computer
for MANY



NUMBER
of devices

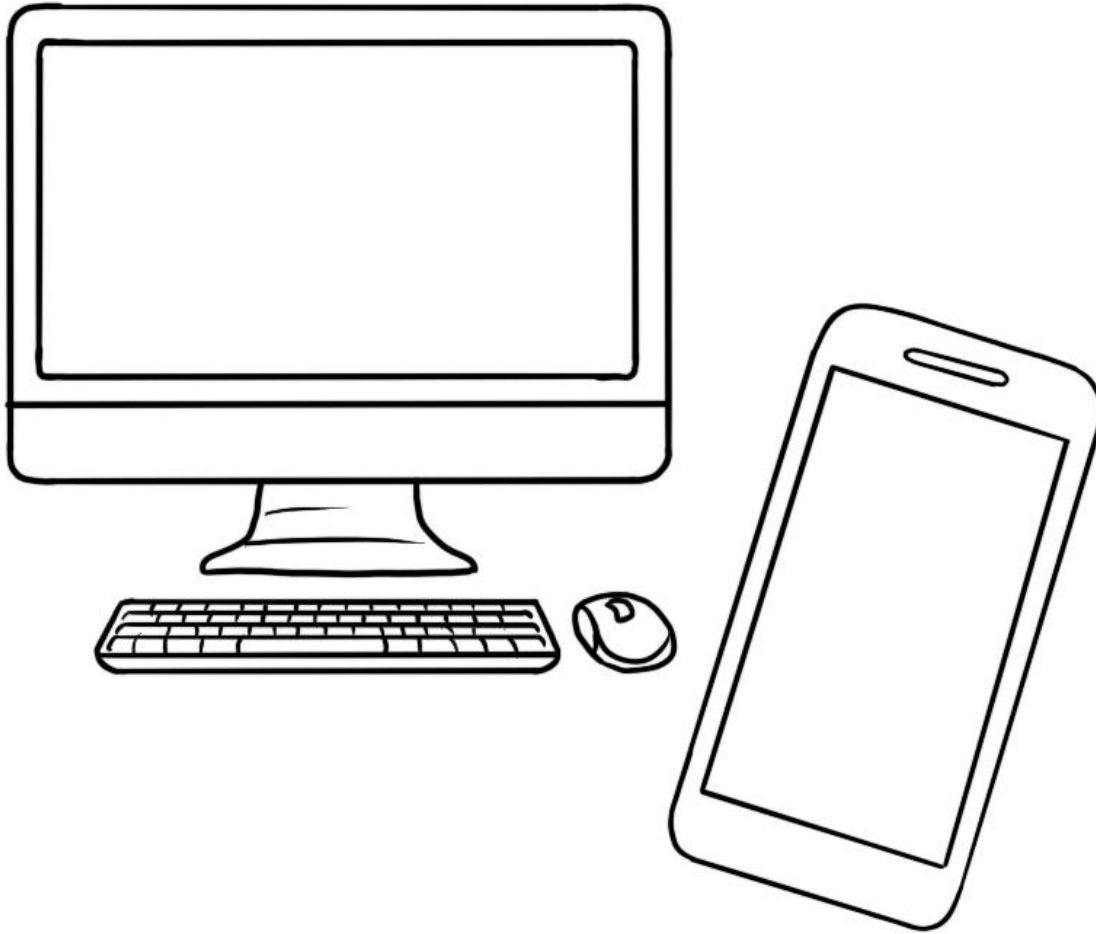
Today, Computing is Ubiquitous.



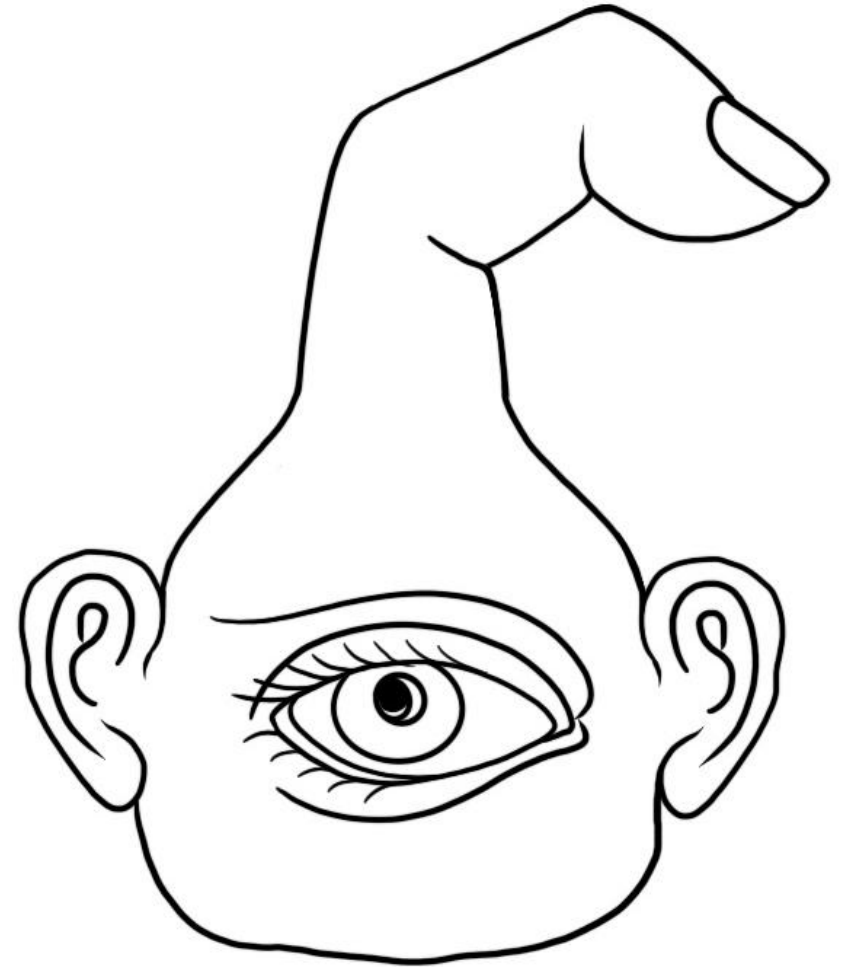
YET, ...

Modern interfaces needed...

How we (most often)
see the computer



How the computer
sees us

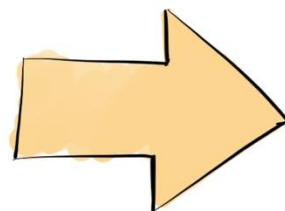


Norman, D. A. (1988). *The Psychology of Everyday Things*. Basic Books.

O'Sullivan, D., & Igoe, T. (2004). *Physical Computing: Sensing and Controlling the Physical World with Computers*. Course Technology Press.

Traditional input devices
are limiting. They are flat,
rigid and disconnected
from the body.





What if we didn't need a
screen to interact - but
could make use of our
whole body instead?



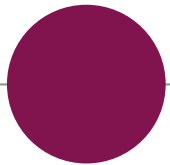
Put-that-There (Bolt, 1980)

- early example of a multimodal interface
- combines *voice* and *gesture* input.
- the output is *visual*.
- demonstrates naturalness and intuitiveness of multimodal interaction



Bolt, R. A. (1980, July). "Put-that-there" Voice and Gesture at the Graphics Interface. In *Proceedings of the 7th annual Conference on Computer Graphics and Interactive Techniques* (pp. 262-270).

Gestural Interfaces



allow us to use specific movements or positions of the hands, fingers, but also the head, gaze or full body used to communicate with a computer.



Computer Museum Paderborn (2020)
Image: Marion Koelle

Common Sensing Approaches

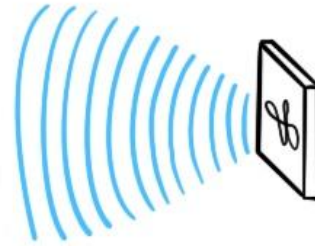
for capturing mid-air, freehand, or even “while grasping” (hand) gestures



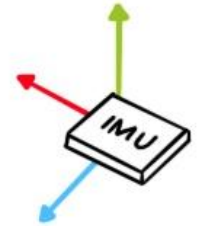
Glove-based



Optical



Radar-based



Inertial
Measurement
Units (IMUs)

Common Sensing Approaches

for capturing mid-air, freehand, or even “while grasping” (hand) gestures



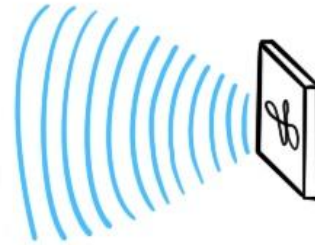
Glove-based

undesirable due
to
bulkiness



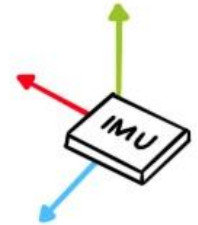
Optical

suffer from
occlusion



Radar-based

limited
field-of-view



Inertial
Measurement
Units (IMUs)

**easy to deploy,
widely used**

Micro-Gesture Dataset

17 IMUs to capture finger movements, 12 everyday objects, 6 grasp types

Freehand



Cylindrical



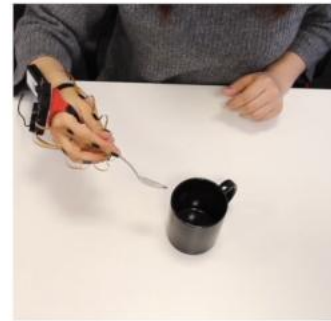
Palmar



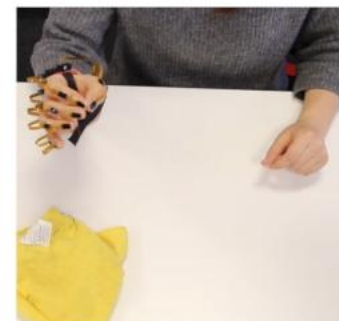
Hook



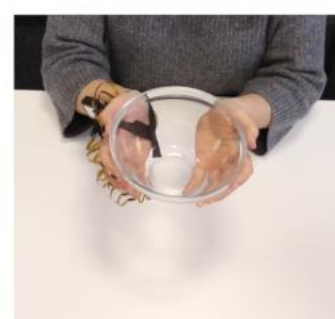
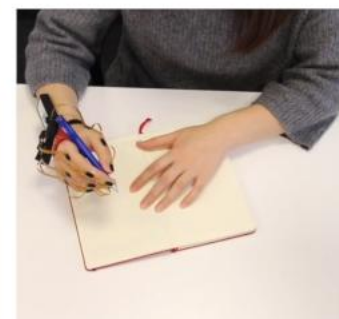
Lateral



Tip

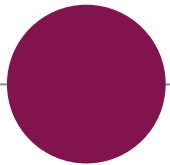


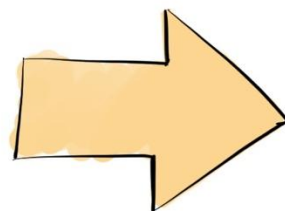
Spherical

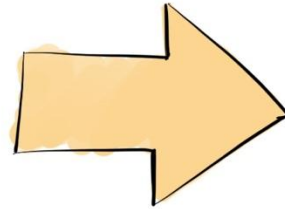


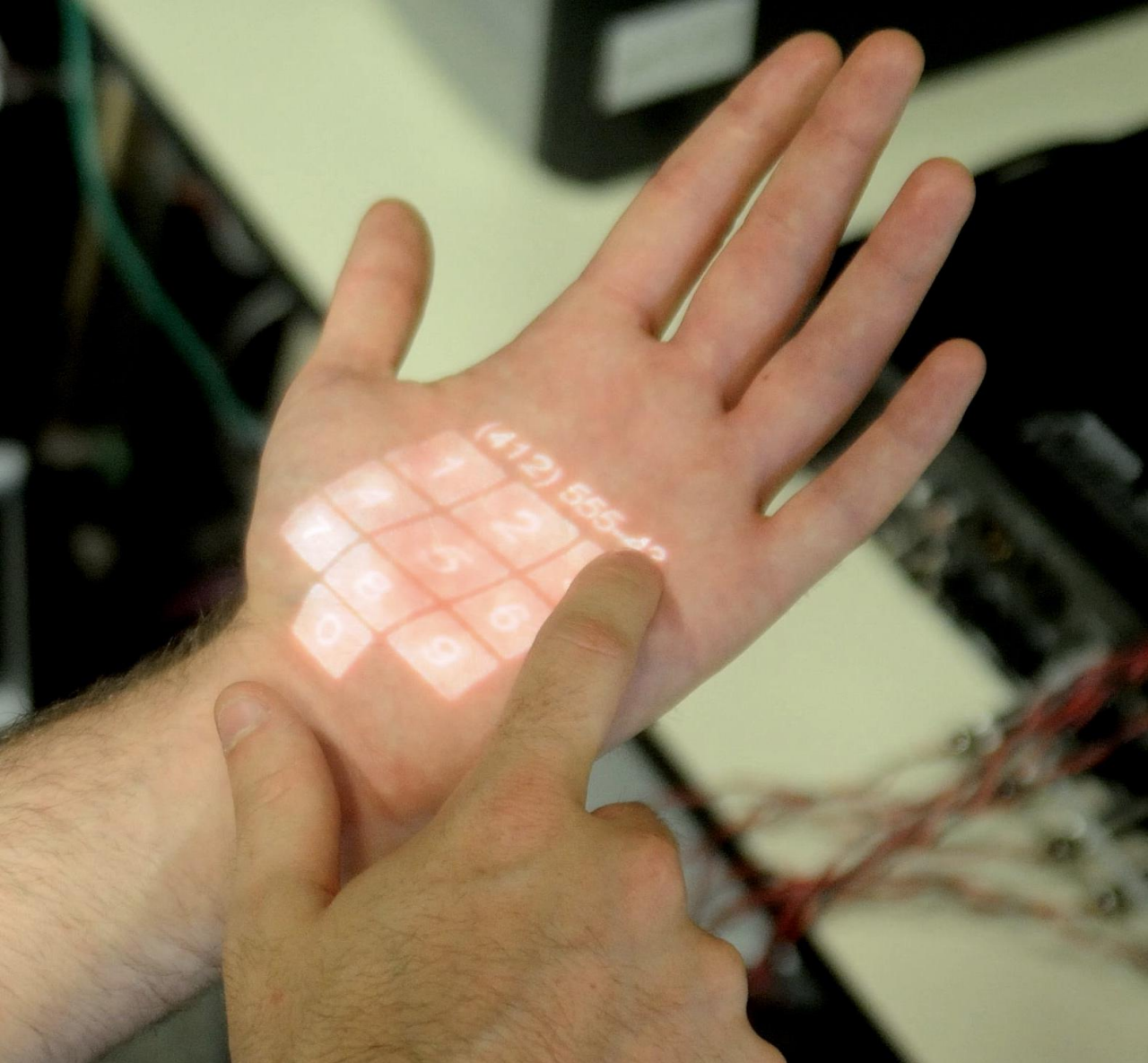
Wearable and On-Body Input

= making use of various electronic devices designed to be worn on the body that are capable of sensing physiological signals and/or explicit user input





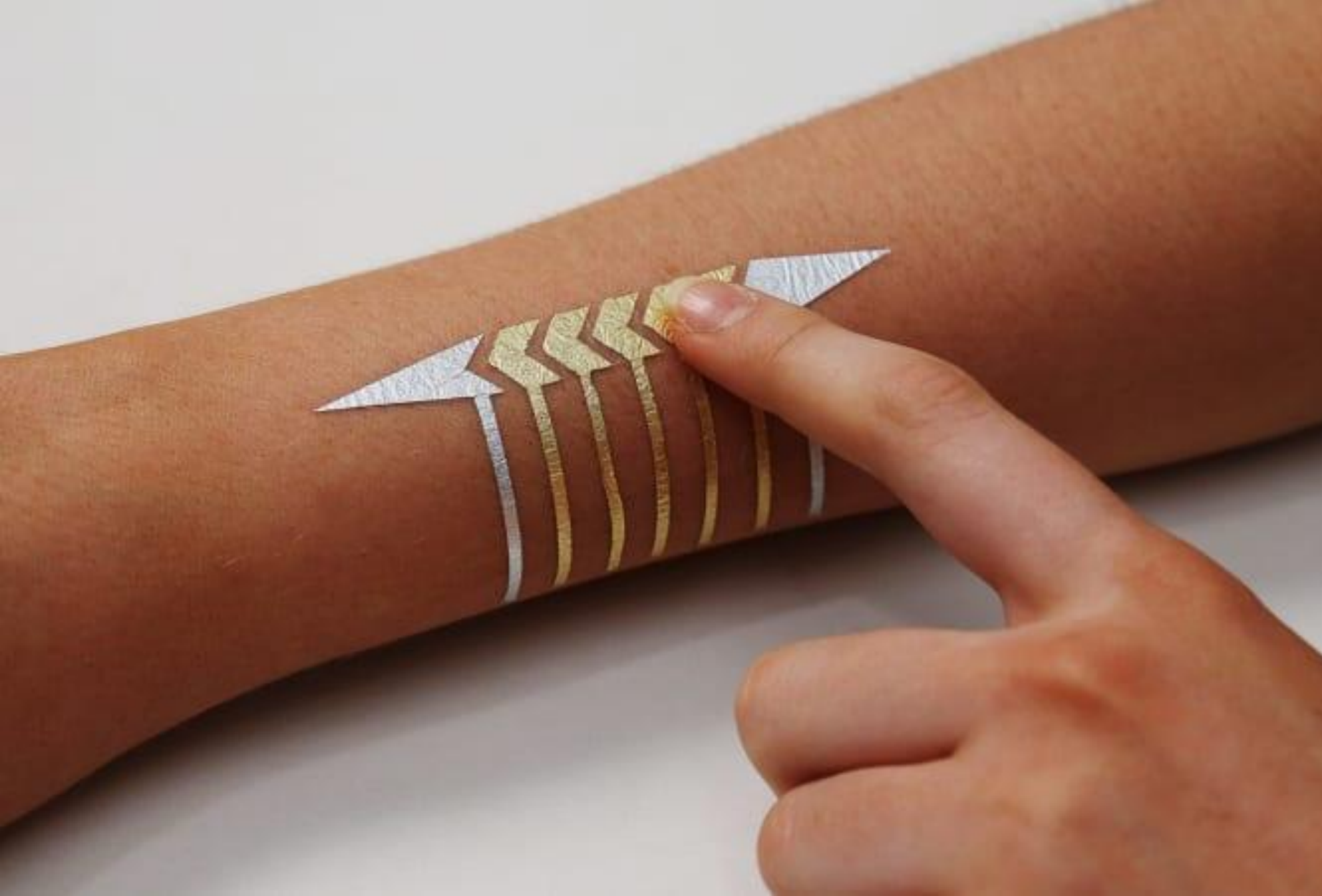




Skinput (Harrison, 2010)

- early work leveraging the skin as input surfaces
- uses a projected interface and an armband sensing micro-vibrations
- various adaptations of the concept, e.g., by He (2025) using a single RGB camera
- patent granted in 2015 to Google LLC

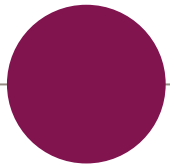






Skinwire (2018)
Kao et al.

Using our Body for Input







Muscle-activated Kinetic Wearables

- textile forms attached to the shoulders of the wearer that (provocatively) expand and contract in response to the movement of the wearer's muscles
- muscle movement is sensed using electrodes attached to the arm
- control was (anecdotally) reported to be intuitive and easy-to-learn



Hartman, K., McConnell, J., Kourtoukov, B., Predko, H., & Colpitts-Campbell, I. (2015). Monarch: Self-Expression Through Wearable Kinetic Textiles.

Image Source: <https://socialbodylab.com/monarch/>, accessed 06/2025

What types of
“Interaction” are
there?

Explicite Interaction

- Explicit interaction focuses on direct interaction between users and devices. The user initiates a direct action and expects a prompt and appropriate response from the device.

Wilson and Oliver, Multimodal Sensing for Explicit and Implicit Interaction, 2005, *HCI International 2005*



Implicite Interaction

- Implicit interaction involves passive, unobtrusive observation of the user over an extended period of time and then responding accordingly.

Wilson and Oliver, Multimodal Sensing for Explicit and Implicit Interaction, 2005, *HCI International 2005*



A customer approaches, is detected, and the sliding door opens automatically without any explicit user interaction.



Multimodal Interaction

When different modalities are used simultaneously in the human-machine interface, this is referred to as ***multimodal interaction***.

Three different types of multimodal interaction:

- ***Complementary***: Avoids redundancy and allows for a natural form of interaction, but can cause problems due to contradictions
- ***Alternative***: Selection of a suitable modality from a range of possible modalities, with some loss of information
- ***Redundant***: Using different modalities at the same time for the same interaction

Multimodal Interaction

Examples

Scenario: Turning on the TV from the comfort of the sofa. Available modalities: *gesture and voice*

Complementary: Point at the TV and say, "On!"
→ Contradictory, since there may be multiple devices in the direction of the TV that can be turned on (stereo, game consoles, etc.)

Alternative: Point at the TV or say, "Turn on the TV!"
→ Only one method; either one would be sufficient on its own to turn on the TV (loss of information: which TV, or what to do)

Redundant: Point at the TV and say, "Turn on the TV!"
→ Different modalities, same information

Current Areas of Research

Ambient Intelligence

HCI in the Field of Ambient Intelligence

Characteristics of an ambient intelligence interaction system:

- A wide variety of *sensors* and *actuators*
- Interaction often—or mostly—occurs *implicitly*
- Systems understand their own functionality, their environment, and the current situation
- Systems perceive their environment through sensors and interpret the data as user behavior

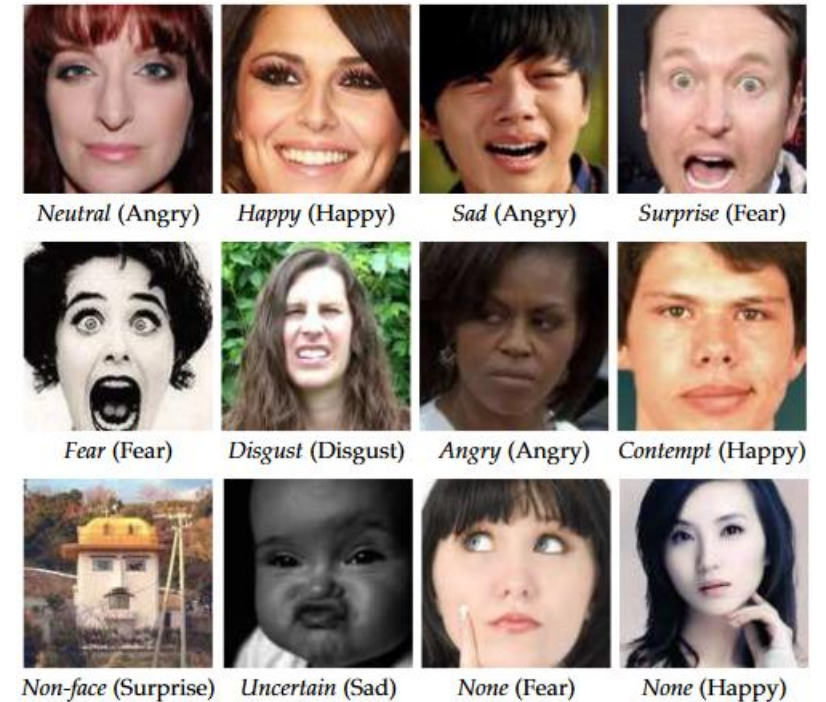
→ Understanding *Context* is required.

Affective Computing

Research and development of systems capable of detecting, interpreting, validating and stimulating human emotions

How can emotions be measured?

Detecting the user's emotions and responding to them



Mollahosseini et al., Affectnet: A database for facial expression, valence, and arousal computing in the wild."
IEEE Transactions on Affective Computing 10.1 (2017)

Affective Computing

Applications:

„Affective Jewelry“:
Earring to measure blood pressure



Galvanic Skin Response (GSR) rings and bracelet



Galvanic Skin Response (GSR) Shoes



Digital Signage

- Interactive displays
- Targeted / Personalised advertisement
- Individualisation/group individualisation based on soft biometric characteristics

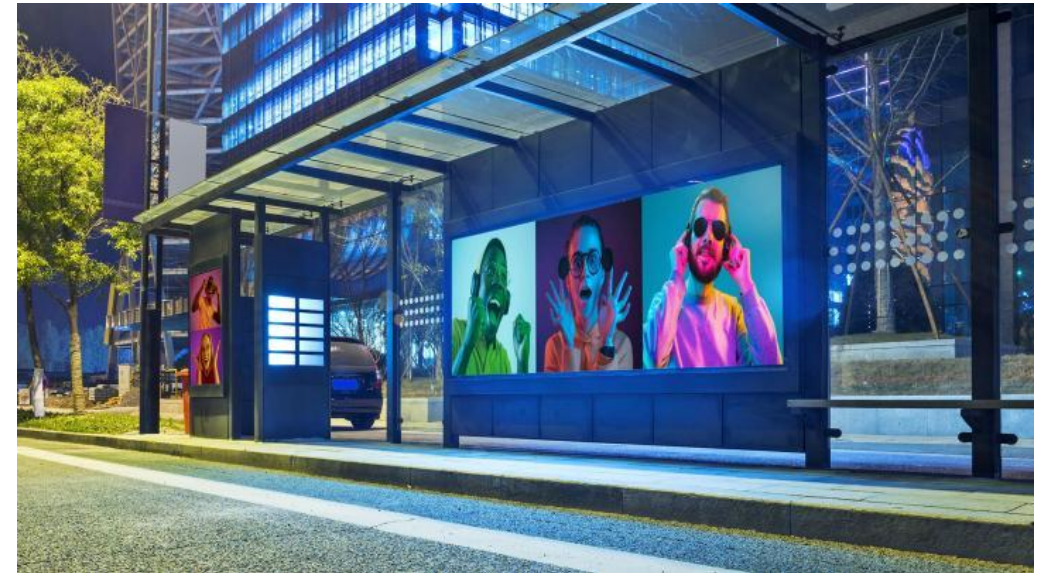
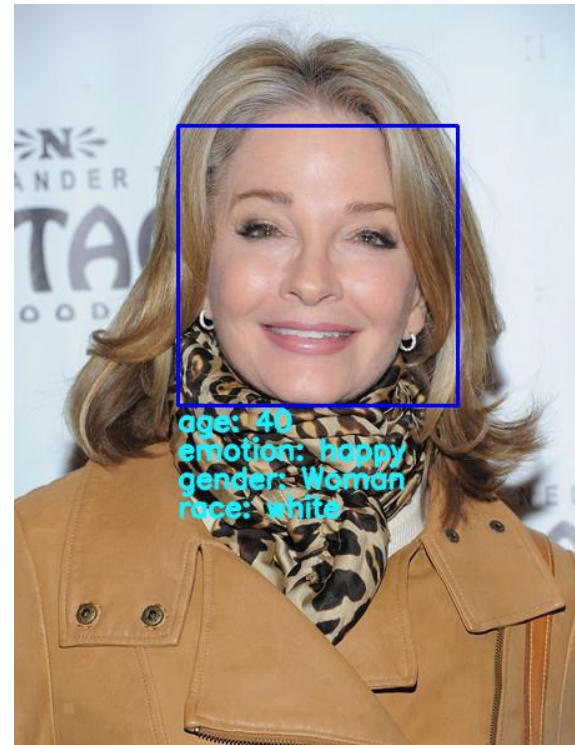
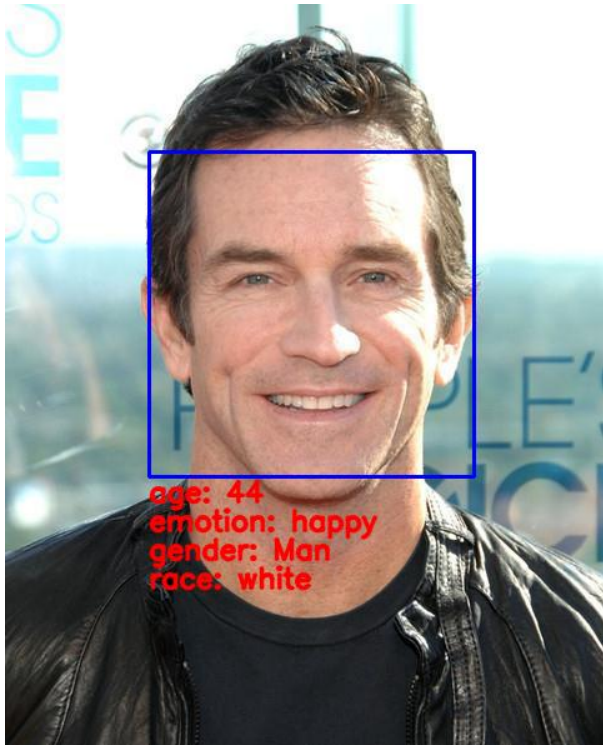


Image: <https://www.intel.de/content/dam/www/public/us/en/images/iot/rwd/a1038512-oooh-digital-billboard-at-bus-stop-at-night-rwd.jpg.rendition.intel.web.720.405.jpg>

Digital Signage

Prediction of soft biometric characteristics from facial images



AR & VR

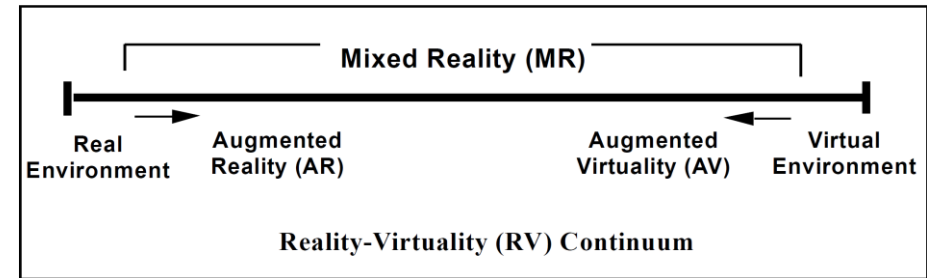
Augmented Reality and Virtual Reality

Virtual Reality: a real or simulated environment in which a person experiences telepresence

Steuer, Jonathan. "Defining virtual reality: Dimensions determining telepresence." *Journal of communication* 42.4 (1992)



Image: <https://visionlib.com/>



Milgram, P., Takemura, H., Utsumi, A., & Kishino, F. (1995, December). Augmented reality: A class of displays on the reality-virtualitycontinuum. In *Photonics for Industrial Applications*(pp. 282-292). International Society for Optics and Photonics.

Machine@Hand



© Fraunhofer IGD

- AR guides users through maintenance and assembly tasks using visual instructions
- AR enables users to see inside the machines
- VR allows users to practise on a virtual model of the machine, even when the physical machine is not on site
- Practical learning for initial and continuing training

Summary

- be familiar with the input & output modalities in HCI
- be able to explain the different types of interaction
- some current research examples in the field of HCI

HCI Research allows us **a glimpse in the future.**

When we create future technologies, we have the **opportunity** and the **responsibility** to design interactive technologies in a way that they **create desirable futures.**

Topics I brought

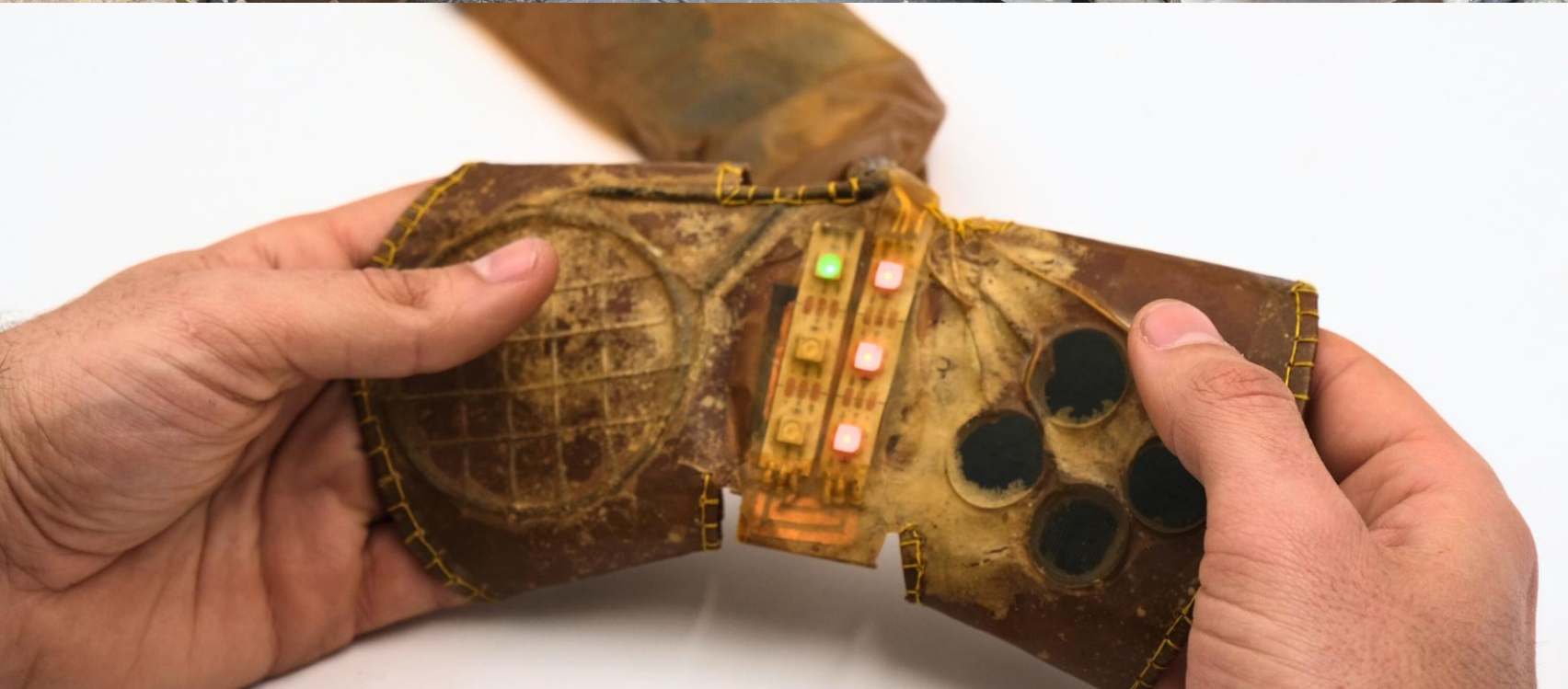
- Cognitive-State-Aware Adaptive Interfaces
- User-Controlled Personalization and Adaptation
- Privacy-Preserving and explainable personalized Interfaces



Prof. Dr. Biying Fu

Explainable AI

Hochschule RheinMain – University
of Applied Sciences and Arts



Reach me via:

biying.fu@hs-rm.de